

Projective varieties with (relative) anti-log canonical divisors being strictly nef

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Background & Motivation

We start from a projective variety $X/\mathbb{C}$.

Definition (Strictly nef)

An $\mathbb{R}$ -Cartier divisor L on X is said to be **strictly nef**, if $L \cdot C > 0$ for every (integral irreducible) curve C on X .

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- (Mumford's example) Let C be a smooth projective curve of genus ≥ 2 . Then there exists a vector bundle E of rank two over C such that the tautological bundle $\mathcal{O}_{\mathbb{P}(E)}(1)$ on the projectivization $\mathbb{P}(E)$ is strictly nef, but not ample.

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Question: How far is a strictly nef divisor from being ample?

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Remark on the positivity of the canonical divisors

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Campana-Peternell's conjecture (1991)

Let X be a smooth projective variety. If $-K_X$ is strictly nef, then X is Fano, i.e., $-K_X$ is ample.

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Theorem (Solution to CP's conjecture in lower dimensions)

CP's conjecture has been confirmed when either $\dim X \leq 2$ (Maeda, 1993) or $\dim X = 3$ (Serrano, 1995).

Background & Motivation

To study the CP's conjecture, Serrano proposed the following.

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Theorem (Partial solution to Serrano's conjecture)

Serrano's conjecture has a positive answer if one of the following holds.

- (i) Gorenstein surfaces, Abelian varieties (Serrano, 1995);
- (ii) $\dim X = 3$ unless X is Calabi-Yau with $L_X \cdot c_2 = 0$ (Campana-Chen-Peternell, 2008; Serrano, 1995); or
- (iii) $\kappa(X) \geq \dim X - 2$ (Campana-Chen-Peternell, 2008).

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- Serrano's conjecture for the case $\dim X \leq n - 1$ is closely related to the proof of CP's conjecture for the case $\dim X = n$.
- Serrano's conjecture for the case $\dim X = n \Rightarrow$ CP's conjecture for the case $\dim X = n$.

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Question (Singular version of Campana-Peternell's conjecture)

Let (X, Δ) be a projective klt pair with the anti-log canonical divisor $-(K_X + \Delta)$ being strictly nef. Is $-(K_X + \Delta)$ ample?

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It has a positive answer on surfaces by studying the generalized pairs.

Theorem (surface case; Han-Liu, 2020)

Let (X, Δ) be a projective lc surface pair, with L_X being a strictly nef Cartier divisor on X . Then $K_X + \Delta + tL_X$ is ample for every real number $t > 3$. In particular, if $-(K_X + \Delta)$ is strictly nef, then $-(K_X + \Delta)$ is ample.

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Theorem (Li-Ou-Yang, 2019)

Let X be a smooth projective variety of dimension n and T_X its tangent bundle. If $\bigwedge^r T_X$ is strictly nef for some $1 \leq r \leq n$, then X is rationally connected. In particular, if $-K_X$ is strictly nef, then X is rationally connected. **A positive answer to the above question in the smooth case.**

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As a corollary, Li-Ou-Yang showed an analogue of Mori's theorem.

Theorem (Li-Ou-Yang, 2019)

A smooth projective variety with strictly nef tangent bundle is $\mathbb{P}^n$.

Background & Motivation

The MRC fibration for klt pair (X, Δ) with nef $-(K_X + \Delta)$ has been recently studied by Matsumura and Wang.

Two extremal cases (two fundamental building blocks):

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- ([Cao, 2016](#)) and ([Cao-Hörling, 2017](#)) showed that the smooth projective varieties with nef anti-canonical divisor are constructed by rationally connected manifolds and Calabi-Yau manifolds.

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In what follows, we recall the result on the MRC fibration for klt pairs with nef anti-log canonical divisors, which is the most significant preliminary ingredient for our project.

Background & Motivation

Theorem (Matsumura-Wang, 2021)

Let (X, Δ) be a projective klt pair with $-(K_X + \Delta)$ being nef. Then, there exists a finite quasi-étale cover $\nu : X' \rightarrow X$ such that:

- (1) X' admits a holomorphic MRC fibration $\psi : X' \rightarrow Y$.
- (2) Y is a projective klt variety with $K_Y \equiv 0$.
- (3) ψ is locally constant with respect to the pair (X', Δ') , where Δ' is the $\mathbb{Q}$ -Weil divisor defined by the pull-back $\Delta' := \nu^* \Delta$. To be precise,
 - (a) For any small open set $B \subseteq Y$, we have $(\psi^{-1}(B), \Delta') \cong B \times (F, \Delta'_F)$ over $B \subseteq Y$, where F is the fibre of ψ and $\Delta'_F := \Delta'|_F$.
 - (b) Every component of Δ' is horizontal.
 - (c) There is a representation $\rho : \pi_1(Y) \rightarrow \text{Aut}(F)$ such that Δ'_F is stable under the action of $\pi_1(Y)$, so that $X' \times_Y Y^{\text{univ}} \cong Y^{\text{univ}} \times F$ and

$$(X', \Delta') \cong (Y^{\text{univ}} \times F, \text{pr}_2^* \Delta'_F) / \pi_1(Y)$$

Background & Motivation

With (X, Δ) replaced by its quasi-étale cover, we get a holomorphic locally constant MRC fibration $(X, \Delta) \xrightarrow{f} Y$ with Y being klt and $K_Y \equiv 0$.

$$\begin{array}{ccc}
 X \times_Y Y^{\text{univ}} \simeq Y^{\text{univ}} \times F & \xrightarrow{f^{\text{univ}} = \text{pr}_1} & Y^{\text{univ}} \\
 \downarrow p_X & \square & \downarrow p_Y \\
 X & \xrightarrow{f} & Y.
 \end{array}$$

- The morphism f is locally trivial analytic fibre bundle with the fibre F .
- Every component Δ_i of Δ is horizontal (i.e., $f(\Delta_i) = Y$).
- There is a representation $\rho : \pi_1(Y) \rightarrow \text{Aut}(F)$ and a Weil $\mathbb{Q}$ -divisor Δ_F on F which is invariant under the action of $\pi_1(Y)$, so that $(X, \Delta) \cong (Y^{\text{univ}} \times F)/\pi_1(Y)$ via the diagonal action.

Main results – Theorem A

First, we give a positive answer to the weak form of the generalized Campana-Peternell conjecture.

Theorem A (Liu-Ou-Wang-Yang-Z., 2021)

Let (X, Δ) be a projective klt pair. Suppose that the anti-log canonical divisor $-(K_X + \Delta)$ is strictly nef. Then X is rationally connected. Moreover, the augmented irregularity $q^\circ(X) = 0$.

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A few remarks on Theorem A:

- $q^\circ(X)$ is defined as the maximum of $q(\tilde{X})$ where $\tilde{X} \rightarrow X$ runs over the finite quasi-étale covers of X .

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- When X is smooth and $\Delta = 0$, it was proved by (Li-Ou-Yang, 2019).
- When $\dim X = 3$, it was proved by (Z., 2021).
- One can further generalize this theorem to the case when $-(K_X + \Delta)$ is **almost strictly nef**, i.e., $-(K_X + \Delta) = \pi^* L_Y$ for some birational morphism $\pi : X \rightarrow Y$ and a strictly nef divisor L_Y on Y .

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Theorem B (Liu-Ou-Wang-Yang-Z., 2021)

Let $f : X \rightarrow Y$ be a fibration from a normal projective variety X to a smooth projective variety Y . Suppose that (X, Δ) is a klt pair and the relative anti-log canonical divisor $-(K_{X/Y} + \Delta)$ is strictly nef. Then f is a locally constant fibration with rationally connected fibres and Y is a canonically polarized hyperbolic projective manifold.

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- Theorem B is a more general form of Theorem A, while Theorem A is one of the key ingredients of the proof of Theorem B.
- We don't know whether the smoothness of Y can be dropped.
- The local constancy of f is essentially proved in (Matsumura-Wang, 21) and (Campana-Cao-Matsumura, 19).

Main results – Theorem C

Theorem C (Liu-Ou-Wang-Yang-Z., 2021)

Let (X, Δ) be a projective klt pair with the anti-log canonical divisor $-(K_X + \Delta)$ being strictly nef. If $\dim X = 3$, then $-(K_X + \Delta)$ is ample.

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- It is still widely open when $\dim X \geq 4$.

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Theorem (Section of the MRC fibration; Liu-Ou-Wang-Yang-Z., 2021)

Let (X, Δ) be a projective klt pair with the anti-log canonical divisor $-(K_X + \Delta)$ being nef. Then up to replacing X with a quasi-étale cover, the MRC fibration $f : X \rightarrow Y$ in [MW21] is a locally constant fibration which admits a section $\sigma : Y \rightarrow X$ such that $\sigma^*(K_X + \Delta) \equiv 0$.

Key ideas for the proof of Theorem A

Theorem (Section of the MRC fibration; Liu-Ou-Wang-Yang-Z., 2021)

Let (X, Δ) be a projective klt pair with the anti-log canonical divisor $-(K_X + \Delta)$ being nef. Then up to replacing X with a quasi-étale cover, the MRC fibration $f : X \rightarrow Y$ in [MW21] is a locally constant fibration which admits a section $\sigma : Y \rightarrow X$ such that $\sigma^*(K_X + \Delta) \equiv 0$.

The above theorem implies our Theorem A:

- We take the (locally constant) MRC fibration $f : X \rightarrow Y$ after replacing X by a quasi-étale cover.
- Suppose the contrary that X is not rationally connected. Then $\dim Y > 0$.
- By the above theorem, up to replacing X by a further quasi-étale cover, our f admits a section $\sigma : Y \rightarrow X$ such that $\sigma^*(K_X + \Delta) \equiv 0$.
- Pick a curve C in Y . By the projection formula,

$$0 = -\sigma^*(K_X + \Delta) \cdot C = -(K_X + \Delta) \cdot \sigma_* C > 0,$$

and we get a contradiction.

Key ideas for the proof of Theorem B

Key ideas for the proof of Theorem B

We start from a fibration $f : (X, \Delta) \rightarrow Y$ of a projective klt pair (X, Δ) onto a smooth projective manifold Y such that $-(K_{X/Y} + \Delta)$ is nef.

Lemma 1

Let T be the null locus of $-(K_{X/Y} + \Delta)$, i.e., the Zariski closure of the union of curves C in X such that $-(K_{X/Y} + \Delta) \cdot C = 0$. If T does not dominate Y , then Y is of general type, and there exists a linear representation $\chi : \pi_1(Y) \rightarrow \mathrm{GL}(r, \mathbb{C})$ whose image is not virtually solvable.

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Lemma 2 (Liu-Ou-Yang, 2020)

Let Y be an irreducible projective variety. If there exists a representation $\rho : \pi_1(Y) \rightarrow \mathrm{GL}(r, \mathbb{C})$ such that its image $\mathrm{im}(\rho)$ is not virtually solvable, then every holomorphic map $\mathbb{C} \rightarrow Y$ is degenerate, i.e., the image is not Zariski dense in Y .

Key ideas for the proof of Theorem B

Theorem (*) (Campana-Cao-Matsumura, 19, Matsumura-Wang, 21)

With the assumption as above, let F be a general fibre of f . Then f is a locally constant fibration with respect to the pair (X, Δ) .

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Now we give the steps to prove our Theorem B.

- By Thm (*), we are left to show Y is hyperbolic and K_Y is ample.
- By Lemma 1, K_Y is big. Therefore, K_Y is ample $\Leftarrow Y$ does not contain any rational curves $\Leftarrow Y$ is hyperbolic.
- Let $h : \mathbb{C} \rightarrow Y$ be holomorphic and $Z := \overline{h(\mathbb{C})}$. Suppose $\dim Z > 0$. Then $h(\mathbb{C})$ is not degenerate in Z . Let $Z' \rightarrow Z$ be the normalization and $f' : X' \rightarrow Z'$ the induced locally constant fibration.
- By Lemma 1, there exists a representation of $\pi_1(Z')$ whose image is not virtually solvable. On the other hand, the holomorphic map h can be lifted to $h' : \mathbb{C} \rightarrow Z'$; thus, $\overline{h'(\mathbb{C})} = Z'$, contradicting Lemma 2.

Key ideas for the proof of Theorem C

Theorem (Han-Liu, 2020)

Let $(X, \Delta + M_X)$ be a 3-dim projective generalized klt pair such that M_X is an $\mathbb{R}_{>0}$ -linear combination of nef Cartier b-divisors. Suppose that $K_X + \Delta + M_X$ is pseudo-effective and $K_X + M_X$ is not pseudo-effective. Then $K_X + \Delta + M_X$ is numerically equivalent to an effective $\mathbb{R}$ -divisor.

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Proposition (Liu-Ou-Yang-Wang-Z., 2021)

Let (X, Δ) be a projective klt threefold pair, and L_X a strictly nef divisor on X . Suppose that L_X is numerically equivalent to an effective $\mathbb{R}$ -divisor. Then $K_X + \Delta + tL_X$ is ample for $t \gg 1$.

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- A klt threefold pair (X, Δ) with $-(K_X + \Delta)$ being strictly nef.
- Take a $\mathbb{Q}$ -factorial terminalization $(Y, \Gamma) \xrightarrow{\tau} (X, \Delta)$ such that (Y, Γ) is a $\mathbb{Q}$ -factorial terminal pair.

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- A klt threefold pair (X, Δ) with $-(K_X + \Delta)$ being strictly nef.
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- If $\Gamma \neq 0$, then we are done by the above two results.

Key ideas for the proof of Theorem C

- So we are left with the case when $\Gamma = 0$. In this case, $\Delta = 0$ and X has only canonical singularities and our theorem follows from (Uehara, 2000). Let us recall the proof here.
- Suppose the contrary that $-K_X$ is not ample.
- $-K_X$ not ample $\Rightarrow -K_X$ not big (base-point-free thm) $\Rightarrow -K_Y$ nef but not big $\Rightarrow K_Y^3 = 0$.
- By a detailed calculation, for a general ample hypersurface S (not contracted by the terminalization τ), the numerical dimension $\nu(Y, -K_Y|_S) = 1$ and the étale fundamental group $\pi_1^{\text{alg}}(S)$ is finite.
- By a theorem of Miyaoka, $-K_Y|_S$ is $\mathbb{Q}$ -linearly equivalent to a non-zero effective divisor.
- In particular, $-K_X|_{\tau(S)}$ is also $\mathbb{Q}$ -linearly equivalent to a divisor $C > 0$ on $\tau(S)$, noting that $-K_X|_{\tau(S)}$ is strictly nef.
- By the projection formula, we get the following contradiction

$$0 = (-K_Y|_S)^2 = (-K_X|_{\tau(S)})^2 = -K_X|_{\tau(S)} \cdot C > 0.$$

Open Question

Question 1

Let $f : (X, \Delta) \rightarrow Y$ be a locally constant fibration with $-(K_{X/Y} + \Delta)$ being **strictly** nef. If we remove the condition that Y is smooth, will we still have f is locally constant?

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Let (X, Δ) be a projective klt pair, and L_X a strictly nef $\mathbb{Q}$ -divisor on X . Is $K_X + \Delta + tL_X$ ample for $t \gg 1$?

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Following the work of Campana-Chen-Peternell, we obtain a partial result on the singular threefold case.

Proposition (Wang-Z., 2021)

Let X be a $\mathbb{Q}$ -factorial Gorenstein terminal projective threefold, and L_X a strictly nef $\mathbb{Q}$ -divisor on X . Suppose that either $\kappa(X) \neq 0$ or $q^\circ(X) \neq 0$. Then $K_X + tL_X$ is ample for $t \gg 1$.

Thank you for your attention!